Type MS · Forming a new quadratic from the roots of another

Answer 1 of 22 · Question 6 part c · 4 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

Number		
6 (a)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *	A1 cso (3)
(b)	$\alpha+\beta=7k$ and $\alpha\beta=k^2$	B1
	$\alpha-\beta=\sqrt{49k^2-4k^2}$	M1
	$=\sqrt{45}k=3k\sqrt{5}$ *	A1 cso (3)
(c)	Sum $=\alpha+\beta\ (=7k)$	B1ft
	Product $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$	M1
	So $x^2-7kx+k^2-3k\sqrt{5}-1=0$	M1A1 (4)
		Total 10 marks

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Part	Mark	Notes
(a)	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	Replaces $\alpha^2+\beta^2$ with $(\alpha+\beta)^2-2\alpha\beta$ in their expansion of $(\alpha-\beta)^2$. And attempts to collect terms $(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$
	A1 cso	For the correct given expression **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
	ALT	
	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	For expanding the RHS and equates to their expansion of $(\alpha-\beta)^2$ and attempting to simplify. $\alpha^2+\beta^2-2\alpha\beta=(\alpha^2+\beta^2+2\alpha\beta)-4\alpha\beta$
	A1 cso	Both sides of the equivalence are shown to be equal **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
(b)	B1	For both $\alpha+\beta=7k$ and $\alpha\beta=k^2$ This may be implied in later work e.g. by use of $49k^2$ and $4k^2$
	M1	For substituting their values for the sum and product into the **given** expression for $(\alpha-\beta)^2$, simplifying and square rooting both sides. $(\alpha-\beta)^2=(7k)^2-4k^2\Rightarrow\alpha-\beta=\sqrt{(7k)^2-4k^2}=\sqrt{45k^2}$ Condone $\pm\sqrt{45k^2}$ for this mark.
	A1 cso	For the correct value of $\alpha-\beta=3k\sqrt{5}$ *
(c)	B1ft	For the sum $(\alpha+1+\beta-1)=\alpha+\beta=7k$ Ft their $\alpha+\beta$
	M1	For the product in terms of k. Correctly multiplying out $(\alpha+1)(\beta-1)$ and substituting in their value $\alpha\beta$ and the correct value $\alpha-\beta=3k\sqrt{5}$ $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$
	M1	For correctly forming an equation with their sum and product $x^2-'7k'x+'k^2-3k\sqrt{5}-1'(=0)$ Condone the absence of = 0 for this mark
	A1	For the correct equation $x^2-7kx+k^2-3k\sqrt{5}-1=0$ Allow $p=-7k\quad q=k^2-3k\sqrt{5}-1$

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Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 2 of 22 · Question 10 part b · 7 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
10(a)	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ $\frac{115}{8}=\left(-\frac{5}{2}\right)^3-3\alpha\beta\left(-\frac{5}{2}\right)$ $\Rightarrow \alpha\beta=\frac{-\frac{115}{8}+\left(-\frac{5}{2}\right)^3}{3\times\left(-\frac{5}{2}\right)}$ $\Rightarrow \alpha\beta=4$	M1 dM1 A1 cso [3]
(b)	**Sum:** $\frac{\alpha^2+1}{\beta}+\frac{\beta^2+1}{\alpha}=\frac{\alpha^3+\beta^3+\alpha+\beta}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}+\frac{\beta^2+1}{\alpha}=\frac{\frac{115}{8}+\left(-\frac{5}{2}\right)}{4}=\frac{95}{32}$ **Product:** $\frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{\alpha^2\beta^2+\alpha^2+\beta^2+1}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{\alpha^2\beta^2+\left[(\alpha+\beta)^2-2\alpha\beta\right]+1}{\alpha\beta}$ $\Rightarrow \frac{\alpha^2+1}{\beta}\times\frac{\beta^2+1}{\alpha}=\frac{4^2+\left(-\frac{5}{2}\right)^2-2\times4+1}{4}=\frac{61}{16}$ **Equation:** $x^2-\left(\frac{95}{32}\right)x+\left(\frac{61}{16}\right)=0\Rightarrow 32x^2-95x+122=0$ oe	M1A1 M1 M1 A1 M1A1 [7]
		Total 10 marks

Type MS · Forming a new quadratic from the roots of another

Answer 2 of 22 · Question 10 part b · 7 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	Part (a) is a 'Show that' question. You must see sufficient work for the award of **both** M marks for the award of the A mark	
	M1	For the correct algebra **and** substitution of the given values into a correct expression for $\alpha^3+\beta^3$ There is more than one acceptable form of this expansion. They must be able to substitute the given values of $\alpha+\beta$ and $\alpha^3+\beta^3$ with $\alpha\beta$ as the value to find in any version they use. For example: $\alpha^3+\beta^3=(\alpha+\beta)\left[(\alpha+\beta)^2-3\alpha\beta\right]$
	dM1	For an attempt to solve the linear equation to find a value for $\alpha\beta$ Allow one processing error for this mark. Note, this is a dependent M mark.
	A1 cso	For $\alpha\beta=4$ This is a show question, you must check their algebra carefully.
(b)	M1	For the **correct** algebra **and** substitution of the given values to find the sum.
	A1	For the correct sum $=\frac{95}{32}$
	M1	For the correct algebra **and** substitution to find $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\left[-\frac{7}{4}\right]$
	M1	For the **correct** algebra **and** substitution of the given values to find the product
	A1	For the correct product $=\frac{61}{16}$
	M1	For forming an equation using **their** sum and product correctly. $x^2-(\text{their sum})x+(\text{their product})=(0)$ Allow missing = 0 for this mark
	A1	For a correct equation with integer coefficients. For example: $64x^2-190x+244=0$

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Answer 2 of 22 · Question 10 part b · 7 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 3 of 22 · Question 5 part b · 6 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5(a)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=$ $\beta^3\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3-3\alpha^2\beta-3\alpha\beta^2=\alpha^3+\beta^3$	M1 A1cso [2]
ALT (a)	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ $=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$ $=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$ $=\alpha^3+\alpha\beta^2-\alpha^2\beta+\beta\alpha^2+\beta^3-\beta^2\alpha$ $=\alpha^3+\beta^3$	 M1 A1 cso [2]
(b)	$\alpha\beta=-\frac{7}{2}$ and $\alpha+\beta=3$ Product: $\frac{\alpha^2}{\beta}\times\frac{\beta^2}{\alpha}=\alpha\beta=-\frac{7}{2}$ Sum: $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\left\{\frac{\alpha^3+\beta^3}{\alpha\beta}\right\}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}=\frac{3^3-3\times-\frac{7}{2}\times3}{-\frac{7}{2}}=-\frac{117}{7}$ Equation: $x^2+\frac{117}{7}x-\frac{7}{2}=0\Rightarrow14x^2+234x-49=0$	B1 B1ft M1A1 M1A1 [6]
		Total 8 marks

Type MS · Forming a new quadratic from the roots of another

Answer 3 of 22 · Question 5 part b · 6 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly expanded $(\alpha+\beta)^3$ $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$
	A1cso	For obtaining the given result without any errors
ALT(a)	M1	Factorises $(\alpha+\beta)$ first then attempts to expand all brackets
	A1cso	For obtaining the given result without any errors
(b)	B1	For the correct sum **and** product of the given equation Allow calling α,β as x_1,x_2 or other letters
	B1ft	For the product $\frac{\alpha^2}{\beta}\times\frac{\beta^2}{\alpha}=\alpha\beta=\text{'}-\frac{7}{2}\text{'}$, follows through their value of $\alpha\beta$
	M1	For the sum $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}=\frac{(\text{'3'})^3-(3)\times\left(\text{'}-\frac{7}{2}\text{'}\right)\times(\text{'3'})}{\left(\text{'}-\frac{7}{2}\text{'}\right)}=\left(-\frac{117}{7}\right)$ Correct algebra and **correct** substitution of values of their $\alpha\beta$ and $\alpha+\beta$
	A1	For the correct sum $-\frac{117}{7}$
	M1	For correctly forming an equation with their sum and product $x^2-\text{sum}\ x+\text{product}=0$ $x^2\text{'}+\frac{117}{7}\text{'}x\text{'}-\frac{7}{2}\text{'}=0$ **Condone the absence of = 0** for this mark.
	A1	For the correct equation $14x^2+234x-49=0$ oe Must be integers coefficients

Type MS · Forming a new quadratic from the roots of another

Answer 3 of 22 · Question 5 part b · 6 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 4 of 22 · Question 11 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

11 (a)	(i) $2\alpha\beta = 3 \Rightarrow q = \alpha\beta \Rightarrow q = \frac{3}{2}$ (ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$, $\left[\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}\right]$ $(\alpha+\beta)^2 = \frac{k^2-6k-3}{4} + 3 = \frac{k^2-6k+9}{4} = \frac{(k-3)^2}{4}$ $\Rightarrow \alpha+\beta = \frac{k-3}{2} \Rightarrow p = \frac{k-3}{2}$	B1 M1A1 M1A1 [5]
(b)	$7\left(\frac{3}{2}\right) = 3\left(\frac{k-3}{2}\right) \Rightarrow 7 = k-3 \Rightarrow k = 10$	M1A1 [2]
(c)	$\alpha+\beta = \frac{10-3}{2} = \frac{7}{2}$ Sum: $\frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta} = \frac{\alpha+\beta}{\alpha+\beta} = 1$ Product: $\frac{\alpha}{\alpha+\beta} \times \frac{\beta}{\alpha+\beta} = \frac{\alpha\beta}{(\alpha+\beta)^2} = \frac{\frac{3}{2}}{\left(\frac{7}{2}\right)^2} = \frac{6}{49}$ Equation: $x^2 - x + \frac{6}{49}$ $(=0)$ $49x^2 - 49x + 6 = 0$	M1 B1 M1 M1 A1 [5]

(a) (i)	B1	$q = \frac{3}{2}$
(ii)	M1	State or use any correct arrangement of $(\alpha+\beta)^2 = \alpha^2+\beta^2+2\alpha\beta$ and substitute $\alpha\beta = \frac{3}{2}$ and $\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}$ oe p may be seen instead of $\alpha+\beta$ at any stage of the working. Condone $-p = \alpha+\beta$ for this mark only.
	A1	Any correct expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$ in terms of k
	M1	Correct method to find an expression for p in terms of k from their expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$. Ignore negative square root. Must identify $p = \alpha+\beta$.
	A1	A correct expression for p in terms of k. The question does not demand simplification, but if it is left as a square root there must be an indication that p must be the positive root (ie no $\pm$ present).

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Answer 4 of 22 · Question 11 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

(b)	M1	Substitute $\alpha\beta = \frac{3}{2}$ and their expression for $\alpha+\beta$ from (a)(ii) into $7\alpha\beta = 3(\alpha+\beta)$ and obtain an equation in k
	A1	$k = 10$ Do not accept if there is an additional root, such as $k = -4$.
(c)	M1	Find a value for $\alpha+\beta$. Need not be simplified. Follow through their k and their expression for p.
	B1	Show or state sum of roots = 1 (stated or explicitly used).
	M1	Write product of roots as $\frac{3/2}{(7/2)^2}$, ft their $\alpha+\beta = \frac{7}{2}$.
	M1	Use equation is $x^2 -$ (sum of roots) + (product of roots) $(=0)$ $(=0$ may be missing). There must be some attempt to substitute values, not just expressions in terms of α and β. May be implied by an equation with fractions correctly cleared.
	A1	$49x^2 - 49x + 6 = 0$ Correct equation with integer coefficients. Must have = 0.

Type MS · Forming a new quadratic from the roots of another

Answer 5 of 22 · Question 7 part b · 8 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	$16-4k(2k-7)\geq 0$	M1
	$2k^2-7k-4\leq 0$	M1
	$(2k+1)(k-4)\leq 0$	M1
	$-\frac{1}{2}\leq k\leq 4$ Accept $-\frac{1}{2}<k<4$	A1 (4)
(b)	$\alpha+\beta=\frac{4}{k}$ $\alpha\beta=\frac{2k-7}{k}$	B1
	$\frac{\alpha+1}{\alpha}+\frac{\beta+1}{\beta}=\frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}=\frac{2\left(\frac{2k-7}{k}\right)+\frac{4}{k}}{\frac{2k-7}{k}}$	M1 M1
	$\frac{4k-10}{k}\times\frac{k}{2k-7}=\frac{2(2k-5)}{2k-7}$	A1
	$\frac{\alpha+1}{\alpha}\times\frac{\beta+1}{\beta}=\frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}=\frac{\frac{2k-7}{k}+\frac{4}{k}+1}{\frac{2k-7}{k}}$	M1 M1
	$\frac{3k-3}{k}\times\frac{k}{2k-7}=\frac{3(k-1)}{2k-7}$	A1
	$(2k-7)x^2-2(2k-5)x+3(k-1)=0$	A1 (8)
	ALTERNATIVE METHOD	
	Let $w=\frac{x+1}{x}$	B1
	$x=\frac{1}{w-1}$	M1
	Hence $k\left(\frac{1}{w-1}\right)^2-4\left(\frac{1}{w-1}\right)+2k-7=0$	M1
	$\frac{k}{(w-1)^2}-\frac{4}{w-1}+2k-7=0$	A1
	$k-4(w-1)+(2k-7)(w-1)^2=0$	M1
	$k-4w+4+(2k-7)(w^2-2w+1)=0$	M1
	$k-4w+4+2kw^2-4kw+2k-7w^2+14w-7=0$	A1
	$(2k-7)x^2-2(2k-5)x+3(k-1)=0$	A1 (8) **[12]**

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	Notes
(a)	
M1	For use of $b^2 - 4ac$ (Ignore inequality for this mark)
M1	For a 3 TQ ≤ 0 oe
M1	For solving their 3TQ (Ignore inequality for this mark) May be implied by $-\frac{1}{2}$ and 4 seen as critical values
A1	For $-\frac{1}{2} \leq k \leq 4$ Allow $<$ instead of $\leq$ Accept $-\frac{1}{2} < k < 4$
(b)	
B1	For $\alpha + \beta = \frac{4}{k}$ and $\alpha\beta = \frac{2k-7}{k}$
M1	For $\frac{\alpha+1}{\alpha} + \frac{\beta+1}{\beta} = \frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}$
M1	For substitution into $\frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}$ with some attempt to simplify
A1	For $\frac{2(2k-5)}{2k-7}$
M1	For $\frac{\alpha+1}{\alpha} \times \frac{\beta+1}{\beta} = \frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$
M1	For substitution into $\frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$ with some attempt to simplify
A1	For $\frac{3(k-1)}{2k-7}$
A1	For $(2k-7)x^2 - 2(2k-5)x + 3(k-1) = 0$
	Alternative
B1	For $w = \frac{x+1}{x}$
M1	For rearranging to make x the subject
M1	For substitution of $x = \frac{1}{w-1}$ into the quadratic
A1	For $\frac{k}{(w-1)^2} - \frac{4}{w-1} + 2k - 7 = 0$
M1	For multiplying by $(w-1)^2$
M1	For expanding brackets
A1	For $k - 4w + 4 + 2kw^2 - 4kw + 2k - 7w^2 + 14w - 7 = 0$
A1	For $(2k-7)x^2 - 2(2k-5)x + 3(k-1) = 0$

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Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 6 of 22 · Question 4 part c · 3 marks

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4 (a)	$\alpha+\beta=-2 \quad \alpha\beta=\frac{3}{2}$	B1
	$(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$	M1
	$(-2)^2-2\left(\frac{3}{2}\right)=4-3=1$ *	M1 A1cso (4)
ALT		
	$2\alpha^2+4\alpha+3=0 \quad$ and $\quad 2\beta^2+4\beta+3=0$	{B1}
	$2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$	{M1}
	$\alpha^2+\beta^2\ =-2\ -2\ -3=1$*	{M1} {A1cso} (4)
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$	M1
	$1\times1-2\left(\frac{3}{2}\right)^2=-\frac{7}{2}$	M1 A1 (3)
ALT		
	$\alpha^2+\beta^2\ ^2=1=\alpha^4+2\alpha^2\beta^2+\beta^4=\alpha^4+\beta^4+2\times\frac{9}{4}$	{M1}
	$\alpha^4+\beta^4=1-\frac{9}{2}=-\frac{7}{2}$	{M1} {A1} (3)
(c)	Product of the roots: $\alpha^4\beta^4=\left(\frac{3}{2}\right)^4=\frac{81}{16}$	M1
	(Sum of the roots: $\alpha^4+\beta^4=-\frac{7}{2}$)	
	$16x^2+56x+81=0$	M1 A1ft (3)
		Total 10 marks

Type MS · Forming a new quadratic from the roots of another

Answer 6 of 22 · Question 4 part c · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\alpha+\beta=-2$ and $\alpha\beta=\frac{3}{2}$
	M1	For $(\alpha+\beta)^2-2\alpha\beta=\alpha^2+\beta^2$ This must be correct
	M1	For substituting their sum and product into their expansion for $\alpha^2+\beta^2$ $(-2)^2-2\left(\frac{3}{2}\right)$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
	ALT	
	B1	For $2\alpha^2+4\alpha+3=0$ and $2\beta^2+4\beta+3=0$
	M1	For $2\ \alpha^2+\beta^2\ +4\ \alpha+\beta\ +6=0$
	M1	For $-2\ -2\ -3$
	A1cso	For obtaining the given expression $\alpha^2+\beta^2=1$
(b)	M1	For $\alpha^4+\beta^4=(\alpha^2+\beta^2)(\alpha^2+\beta^2)-2\alpha^2\beta^2$ This must be correct
	M1	For substituting $\alpha^2+\beta^2=1$ and their product from (a) into their expansion for $\alpha^4+\beta^4$ $\alpha^4+\beta^4=1\times1-2\left(\frac{3}{2}\right)^2$
	A1	For the correct value $\alpha^4+\beta^4=-\frac{7}{2}$
	ALT	
	M1	For $1=\alpha^4+2\alpha^2\beta^2+\beta^4$
	M1	For $1=\alpha^4+\beta^4+2\times\frac{9}{4}$
	A1	For $\alpha^4+\beta^4=-\frac{7}{2}$
(d)	M1	For product of the roots: $\alpha^4\beta^4\left(=\left(\frac{3}{2}\right)^4=\frac{81}{16}\right)$
	M1	For use of $x^2-(\text{their sum})\,x+(\text{their product})=[0]$
	A1	For $16x^2+56x+81=0$ Must include = 0

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Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 7 of 22 · Question 8 part c · 6 marks

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8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	$(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2\Rightarrow\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	$\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$	M1
	$k^4+12k^2-448=(k^2-16)(k^2+28)=0\Rightarrow k^2=16\Rightarrow k=4$	M1A1 [5]
(c)	Sum: $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	$\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	Product: $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	$\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	3TQ: $x^2+\frac{412}{27}x-\frac{436}{27}=0\Rightarrow 27x^2+412x-436=0$	M1A1 [6]
		Total 14 marks

Question	Scheme	Marks
8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	For the correct algebra on $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	For substituting *their* sum and *their* product to give a simplified expression for $\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]

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Answer 7 of 22 · Question 8 part c · 6 marks

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(b)	For the correct algebra on $\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2$ Or $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2$	M1
	For substituting the values for $\alpha^2+\beta^2$ (in terms of k) and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$ $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$ Or for substituting the values for $\alpha^2+\beta^2$ (in terms of k), $\alpha+\beta$ and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$	M1
	For forming a simplified quadratic. $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$ Or for $\left(\frac{k^2+6}{9}\right)^2=\frac{448}{81}$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For solving their quadratic. $k^4+12k^2-448=\left(k^2-16\right)\left(k^2+28\right)=0\Rightarrow k^2=16\Rightarrow k=4$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For the correct value of $k=4$ If they give $k=\pm4$ this is A0	A1 [5]
(c)	For the correct algebra on the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	For the correct algebra on the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	For the 3TQ from their values of the sum and product (used correctly) [= 0 not required for this mark].	M1

Type MS · Forming a new quadratic from the roots of another

Answer 7 of 22 · Question 8 part c · 6 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

	$x^2+\frac{412}{27}x-\frac{436}{27}=(0)$	
	For the correct 3TQ. Must have $= 0$. $27x^2+412x-436=0$ Must have integer coefficients. Accept a multiple of this.	A1 [6]
		Total 14 marks

Question	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 8 of 22 · Question 10 part b · 4 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

10(a)	$\alpha+\beta=-\frac{k}{2}\qquad \alpha\beta=2$	B1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta,\quad \alpha^2-\beta^2=(\alpha+\beta)(\alpha-\beta)$	B1B1
	$\Rightarrow \alpha^2-\beta^2=(\alpha+\beta)\sqrt{(\alpha+\beta)^2-4\alpha\beta}$	
	$\Rightarrow \alpha^2-\beta^2=\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times 2}=\frac{7\sqrt{17}}{4}$	M1
	$\Rightarrow \frac{k^4}{4}-8k^2-\frac{833}{4}=0\Rightarrow k^4-32k^2-833=0$ or any equivalent	M1A1
	$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7$ *	M1A1 cso [8]
(b)	Product: $\left[(\alpha-\beta)(\alpha+\beta)=\alpha^2-\beta^2=\frac{7\sqrt{17}}{4}\right]$ from (a)	B1
	Sum: $(\alpha-\beta)+(\alpha+\beta)=\frac{\sqrt{17}}{2}+\frac{7}{2}=\left(\frac{\sqrt{17}+7}{2}\right)$	M1
	$x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0\Rightarrow 4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$	M1A1 [4]
		Total 12 marks

Type MS · Forming a new quadratic from the roots of another

Answer 9 of 22 · Question 6 part a · 6 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

6(a)	$\alpha+\beta=\frac{3}{4}\quad \alpha\beta=-\frac{5}{4}$	B1
	$\frac{2\alpha}{\beta}+\frac{2\beta}{\alpha}=\frac{2\left(\alpha^2+\beta^2\right)}{\alpha\beta}=\frac{2\left(\left(\alpha+\beta\right)^2-2\alpha\beta\right)}{\alpha\beta}$	M1
	$=\frac{2\left(\frac{9}{16}+\frac{5}{2}\right)}{-\frac{5}{4}}=-\frac{49}{10}$	A1
	$\frac{2\alpha}{\beta}\times\frac{2\beta}{\alpha}=4$	B1
	$x^2+\frac{49}{10}x+4\ (=0)$	M1
	$10x^2+49x+40=0$	A1 (6)
(b)	$4\left(\alpha+\beta\right)=3=-\frac{p}{4}\quad p=-12$	M1A1
	$\left(3\alpha+\beta\right)\times\left(\alpha+3\beta\right)=3\left(\alpha^2+\beta^2\right)+10\alpha\beta$	M1
	$=3\left(\frac{9}{16}+\frac{5}{2}\right)-10\times\left(\frac{5}{4}\right)=-\frac{53}{16}$ oe	A1
	$\frac{q}{4}=-\frac{53}{16}\quad q=-\frac{53}{4}$ oe	A1 (5)
		[11]

(a)B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
M1	Attempt the sum of the roots of the new equation. Must reach a **correct** expression ready for substitution of values of $\alpha+\beta$ **and** $\alpha\beta$
A1	Correct value for the sum. Allow if $\alpha+\beta=-\frac{3}{4}$ has been used.
B1	Correct value for the product of roots of the new equation.
M1	Use $x^2-\left(\text{sum of roots}\right)x+\text{product of roots}$ "= 0" may be missing
A1	Correct final equation, as shown or an **integer** multiple of this. Must have = 0 now. Allow if $\alpha+\beta=-\frac{3}{4}$ has been used.
(b)	
M1	Use their value of $\alpha+\beta$ to obtain a value for the sum of the roots of $g\left(x\right)=0$ and equate to $\pm p/4$
A1	Obtain the correct value of p
M1	Attempt the product of the roots of $g\left(x\right)=0$ and obtain an expression ready for substitution of values. May use work from (a) for value of $\alpha^2+\beta^2$ so the expression shown is sufficient.
A1	Obtain the correct value for the product
A1	Correct value of q

Type MS · Forming a new quadratic from the roots of another

Answer 9 of 22 · Question 6 part a · 6 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$(x+1)\quad(4x-2)$	

Type MS · Forming a new quadratic from the roots of another

Answer 10 of 22 · Question 8 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8(a)	$x^3+3x^2y+3xy^2+y^3$	B1 [1]
(b)	$\alpha+\beta=-\frac{3}{2}$ **and** $\alpha\beta=\frac{4}{2}$ oe	B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$\left("-\frac{3}{2}"\right)^3-3("2")\left("-\frac{3}{2}"\right)$ oe	dM1
	$\frac{45}{8}$ oe	A1 [4]
(c)	$\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=$ $\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}=\frac{\left("\frac{45}{8}"\right)}{"2"^2}$	M1
	$\frac{45}{32}$ oe	A1ft
	$\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ $\frac{1}{2}$ oe	B1ft
	$x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$	M1
	$32x^2-45x+16=0$ oe	A1 [5]
		Total 10 marks

Type MS · Forming a new quadratic from the roots of another

Answer 10 of 22 · Question 8 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the correct expansion simplified or un-simplified. For example, allow $x^3+2x^2y+x^2y+xy^2+2xy^2+y^3$
(b)	B1	Product and sum both correct, written explicitly or used later in working.
	M1	An attempt to rearrange their expression from part (a) to achieve as a minimum $\alpha^3+\beta^3=(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** Accept alternative algebraic arrangements of $\alpha^3+\beta^3$ but please check carefully that the algebra is correct.
	dM1	Substitution of **their** sum and product into an expression which must be of the form $(\alpha+\beta)^3\pm3\alpha\beta(\alpha+\beta)$ **Note:** If they substitute correctly into alternative arrangements check that they are correct.
	A1	For $\frac{45}{8}$ oe
(c)	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	Use of their sum and product to correctly form an expression as shown, $x^2-"\frac{45}{32}"x+"\frac{1}{2}"(=0)$ Allow missing = 0 for this mark or even = y for this mark.
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.
	ALT	Starts with $\left(x-\frac{\alpha}{\beta^2}\right)\left(x-\frac{\beta}{\alpha^2}\right)=0\Rightarrow x^2-x\left(\frac{\beta}{\alpha^2}+\frac{\alpha}{\beta^2}\right)+\frac{\beta}{\alpha^2}\times\frac{\alpha}{\beta^2}=0$
	M1	Correct algebra to obtain $\frac{\alpha}{\beta^2}+\frac{\beta}{\alpha^2}=\frac{\alpha^3+\beta^3}{\alpha^2\beta^2}$ **and** substitution of **their** values for the sum and product.
	A1ft	For $\frac{45}{32}$ oe **Note:** only follow through their values for product and sum into **correct algebra.**
	B1ft	For the correct algebra and substitution of their $\alpha\beta=2$ to obtain $\frac{\alpha}{\beta^2}\times\frac{\beta}{\alpha^2}=\frac{1}{\alpha\beta}=\frac{1}{'2'}$ ft their $\alpha\beta$
	M1	For $x^2-x\left('\frac{45}{32}'\right)+'\frac{1}{2}'=0$ Accept this expression on it's own or even = y for this mark
	A1	Correct equation or any multiple of it, for example , $64x^2-90x+32=0$ The coefficients, must be integers.

Type MS · Forming a new quadratic from the roots of another

Answer 10 of 22 · Question 8 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 11 of 22 · Question 2 part whole · 8 marks

2024 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

	ALT 2 – Polynomial division	
	M1	For obtaining as a minimum $a - k = 15$
	A1	For obtaining $a = 21$ with no errors seen
(b)	**Polynomial division**	
	M1	Uses polynomial division to achieve as a minimum $2x^2 \pm 3x \pm k$ where k is a constant **NB** Award this seen in part (a) if they use polynomial division
	A1	For the correct quadratic factor $2x^2 - 3x + 5$
	OR Equates coefficients	
	M1	Achieves the correct values for a and b
	A1	For the correct quadratic factor $2x^2 - 3x + 5$
	Uses the discriminant	
	M1	Finds the value of the discriminant using their QE Allow the discriminant to be embedded as part of the Quadratic formula. If they do not achieve the correct 3TQ – please check their discriminant carefully. **NB: It must be evaluated,**
	A1	For a value of –31 for the discriminant with a conclusion. Minimally acceptable conclusion: • Discriminant < 0 hence no real roots. • The only intersection [root/solution] is at $x = \frac{2}{3}$
	Completes square	
	M1	Completes the square on their 3TQ as far as: $\left(x - \frac{3}{4}\right)^2 = -\frac{31}{16}$
	A1	For a value of $-\frac{31}{16}$ with a conclusion • $-\frac{31}{16} < 0$ so the quadratic has no real roots • The only intersection [root/solution] with the x-axis is at $x = \frac{2}{3}$
	Complex roots	
	The question states clearly 'Show Algebraically' so just quoting the complex roots is M0. If they go through the process of finding them fully, please send to REVIEW	

Type MS · Forming a new quadratic from the roots of another

Answer 11 of 22 · Question 2 part whole · 8 marks

2024 JUN · Paper 02 · extract 2 of 5 · find part whole in this mark scheme extract

Question	Scheme	Marks
2	$\alpha+\beta=-\left(-\frac{5}{3}\right)=\frac{5}{3},\quad \alpha\beta=\frac{1}{3}$	B1,B1
	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\Rightarrow\left(\frac{5}{3}\right)^2-2\times\frac{1}{3}=\frac{19}{9}$	B1
	PRODUCT: $\frac{\alpha}{2\beta}\times\frac{\beta}{2\alpha}=\frac{1}{4}$	B1
	SUM: $\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{2\alpha^2+2\beta^2}{4\alpha\beta}=\left[\frac{\alpha^2+\beta^2}{2\alpha\beta}\right]\Rightarrow\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{\frac{19}{9}}{\frac{2}{3}}=\frac{19}{6}$	M1A1
	EQUATION: $x^2-\frac{19}{6}x+\frac{1}{4}=0\Rightarrow 12x^2-38x+3=0$ o.e.	M1A1 [8]
		Total 8 marks

Type MS · Forming a new quadratic from the roots of another

Answer 11 of 22 · Question 2 part whole · 8 marks

2024 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Question	Scheme	Marks
2	$\alpha+\beta=-\left(-\frac{5}{3}\right)=\frac{5}{3},\quad \alpha\beta=\frac{1}{3}$	B1,B1
	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\Rightarrow\left(\frac{5}{3}\right)^2-2\times\frac{1}{3}=\frac{19}{9}$	B1
	PRODUCT: $\frac{\alpha}{2\beta}\times\frac{\beta}{2\alpha}=\frac{1}{4}$	B1
	SUM: $\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{2\alpha^2+2\beta^2}{4\alpha\beta}=\left[\frac{\alpha^2+\beta^2}{2\alpha\beta}\right]\Rightarrow\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{\frac{19}{9}}{\frac{2}{3}}=\frac{19}{6}$	M1A1
	EQUATION: $x^2-\frac{19}{6}x+\frac{1}{4}=0\Rightarrow 12x^2-38x+3=0$ o.e.	M1A1 [8]
		Total 8 marks

Type MS · Forming a new quadratic from the roots of another

Answer 11 of 22 · Question 2 part whole · 8 marks

2024 JUN · Paper 02 · extract 4 of 5 · find part whole in this mark scheme extract

Mark	Notes
B1	For the correct sum of $\frac{5}{3}$ This may not be explicitly seen – look for it in their working
B1	For the correct product of $\frac{1}{3}$ This may not be explicitly seen – look for it in their working
B1	For the correct value of $\alpha^2+\beta^2$ $\left[\frac{19}{9}\right]$ but does not need to be evaluated so accept $\left[\left(\frac{5}{3}\right)^2-2\times\frac{1}{3}\right]$ seen anywhere in their working. This can be embedded anywhere in their work This can also be implied from a correct value of the SUM $\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{19}{6}$
B1	For the correct PRODUCT of $\frac{1}{4}$
M1	For the correct algebra $\left[\frac{2\alpha^2+2\beta^2}{4\alpha\beta}\right]$ **and** substitution of their values of $\alpha^2+\beta^2$ [even if the algebra on $\alpha^2+\beta^2$ is incorrect] to find the SUM. Ft their $\frac{19}{9}$ even if it comes from incorrect algebra on $\alpha^2+\beta^2$ earlier.
A1	For the value of $\frac{19}{6}$
M1	For a correct equation in any form. Need not = 0 for this mark
A1	For a correct equation with integer coefficients = 0

Type MS · Forming a new quadratic from the roots of another

Answer 11 of 22 · Question 2 part whole · 8 marks

2024 JUN · Paper 02 · extract 5 of 5 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 12 of 22 · Question 5 part d · 6 marks

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5 (a)	$(\alpha+\beta)^3 = \alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3 \Rightarrow \alpha^3+\beta^3$ $=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$*	M1A1 cso [2]
(b)	$\alpha+\beta=-\frac{3}{2} \qquad \alpha\beta=\frac{6}{2}=3$ $\left[\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right]$ $\alpha^3+\beta^3=\left(-\frac{3}{2}\right)^3-3\times3\times\left(-\frac{3}{2}\right)=\frac{81}{8}$	B1 B1 [2]
(c)	$\left(\alpha^2+\beta^2\right)^2=\alpha^4+2\alpha^2\beta^2+\beta^4 \Rightarrow \alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2$*	M1A1 cso [2]
(d)	$\alpha^2+\beta^2=\left(-\frac{3}{2}\right)^2-2\times3=-\frac{15}{4}$ Sum $\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\alpha^3+\beta^3-(\alpha+\beta)=\frac{81}{8}-\left(-\frac{3}{2}\right)=\frac{93}{8}$ Product $\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$ $=27-\left[\left(-\frac{15}{4}\right)^2-2\times3^2\right]+3=\frac{543}{16}$ Equation $x^2-\frac{93}{8}x+\frac{543}{16}=0 \Rightarrow 16x^2-186x+543=0$	B1 B1 M1 A1 M1A1 [6]
		Total 12 marks

Type MS · Forming a new quadratic from the roots of another

Answer 12 of 22 · Question 5 part d · 6 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

(d)	
B1	$\alpha^2+\beta^2=-\frac{15}{4}$ $\left(\text{May be implied by } \alpha^4+\beta^4=-\frac{63}{16}\right)$
B1	$\left(\alpha^3-\beta\right)+\left(\beta^3-\alpha\right)=\frac{93}{8}$
M1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\left(\alpha\beta\right)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$
A1	$\left(\alpha^3-\beta\right)\times\left(\beta^3-\alpha\right)=\frac{543}{16}$
M1	$x^2-\text{'sum'}\,x+\text{'product'}\left(=0\right)$
A1	$16x^2-186x+543=0$

Type MS · Forming a new quadratic from the roots of another

Answer 12 of 22 · Question 5 part d · 6 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 13 of 22 · Question 1 part whole · 7 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

4. Use of calculators

Unless the question specifically states 'show' or 'prove' accept correct answers from no working. If an incorrect solution is given without any working do not award the Method mark.

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Forming a new quadratic from the roots of another

Answer 13 of 22 · Question 1 part whole · 7 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this: eg in a case of "prove or show....")

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Multiple attempts at a question.

If there is a choice of methods shown, then award the lowest mark, unless the subsequent working makes clear the method that has been used.

Type MS · Forming a new quadratic from the roots of another

Answer 13 of 22 · Question 1 part whole · 7 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks
1	$\alpha+\beta=\frac{3}{4},\ \alpha\beta=-2$	B1B1
	Sum: $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{\frac{3}{4}}{-2}=-\frac{3}{8}=\left(-\frac{b}{a}\right)$	M1A1
	Product: $\frac{1}{\alpha}\times\frac{1}{\beta}=\frac{1}{\alpha\beta}=-\frac{1}{2}=\left(\frac{c}{a}\right)$	B1ft
	Equation: $x^2+\frac{3}{8}x+\left(-\frac{1}{2}\right)=0\Rightarrow 8x^2+3x-4=0$	M1A1ft [7]
		Total 7 marks

Type MS · Forming a new quadratic from the roots of another

Answer 14 of 22 · Question 6 part b · 5 marks

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6(a)	$\alpha+\beta=-\frac{5}{2}\quad \alpha\beta=-\frac{p}{2}$	B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$\alpha^3+\beta^3=\left(-\frac{5}{2}\right)^3-3\left(-\frac{p}{2}\right)\left(-\frac{5}{2}\right)=-\frac{125}{8}-\frac{15p}{4}=-\frac{215}{8}$	M1M1
	$\Rightarrow p=3$	A1
	ALT	[5]
	$\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$	[M1]
(b)	Sum:	
	$\frac{\alpha+\beta}{\alpha^2}+\frac{\alpha+\beta}{\beta^2}=\frac{\alpha\beta^2+\beta^3+\alpha^3+\alpha^2\beta}{\alpha^2\beta^2}=\frac{\alpha^3+\beta^3+\alpha\beta(\alpha+\beta)}{\alpha^2\beta^2}$	M1
	ALT	
	$\frac{\alpha+\beta}{\alpha^2}+\frac{\alpha+\beta}{\beta^2}=\frac{\alpha^2(\alpha+\beta)+\beta^2(\alpha+\beta)}{\alpha^2\beta^2}=\frac{(\alpha^2+\beta^2)(\alpha+\beta)}{\alpha^2\beta^2}$	M1
	$=\frac{\left((\alpha+\beta)^2-2\alpha\beta\right)(\alpha+\beta)}{\alpha^2\beta^2}$	
	$\frac{-\frac{215}{8}+\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{\left(-\frac{3}{2}\right)^2}=-\frac{185}{18}$ allow $\frac{185}{18}$ if they start with a negative.	B1
	Product:	
	$\left(\frac{\alpha+\beta}{\alpha^2}\right)\times\left(\frac{\alpha+\beta}{\beta^2}\right)=\frac{(\alpha+\beta)^2}{\alpha^2\beta^2}=\frac{\left(-\frac{5}{2}\right)^2}{\left(-\frac{3}{2}\right)^2}=\frac{25}{9}$	M1A1 [5]
	Equation:	
	$x^2+\frac{185}{18}x+\frac{25}{9}=0\Rightarrow 18x^2+185x+50=0$ oe	
		Total 10 marks

Type MS · Forming a new quadratic from the roots of another

Answer 14 of 22 · Question 6 part b · 5 marks

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Part	Mark	Notes
(a)	B1	For the correct expression/values for **BOTH** the sum and product. This must be identified, or implied from their working.
	M1	For the correct algebra to find $\alpha^3+\beta^3$ in terms of $\alpha+\beta$ and $\alpha\beta$ • $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ • $\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$ • Or any other algebra, but do not award this mark until the values of $\alpha+\beta$ and $\alpha\beta$ can be substituted in directly.
	M1	For substituting their values of the sum and product into their expression for $\alpha^3+\beta^3$
	M1	For solving the **linear** equation in p Allow one slip in their working.
	A1	For the correct value of p with no errors.
(b)	M1	For the correct algebra for the sum of roots. This must be such that the given value of $\alpha^3+\beta^3$, with their values of $\alpha\beta$ and $(\alpha+\beta)$ can be substituted in. If they use the **ALT** they will not need $\alpha^3+\beta^3$ Some candidates will reverse the sign at this stage in anticipation of the reversal required in the equation.
	M1	For substituting in the given value for $\alpha^3+\beta^3$, and their values for $\alpha+\beta$ and $\alpha\beta$ into their expansion for the sum. If they use the **ALT** they will not need $\alpha^3+\beta^3$
	B1ft	For the correct value of the product of roots. You must follow through their p. You do not need to check the calculation once you see the correct algebra with a correct substitution.
	M1	For forming an equation with their sum and product. Ft the sign of their sum. Some candidates reverse the sign when finding the sum. Watch out for that! Accept this without = 0
	A1	For a correct equation including = 0 There is no follow through on this mark. **NB** accept a correct equivalent equation provided it has integer coefficients. e.g. $36x^2+370x+100=0$
	SC – they solve the equation and find values for α and β Award marks for correct work seen above in part (b) • If they do not obtain an expansion for the sum into which $\alpha^3+\beta^3$, $\alpha\beta$ and $(\alpha+\beta)$ cannot be directly substituted – first M0 • If they do not substitute $\alpha^3+\beta^3$, $\alpha\beta$ and $(\alpha+\beta)$ but other values based on α and β - second M0 • If they cannot substitute $\alpha\beta$ and $(\alpha+\beta)$ into the product – B0 Then allow marks for forming the equation as above.	

Type MS · Forming a new quadratic from the roots of another

Answer 14 of 22 · Question 6 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 15 of 22 · Question 10 part c · 6 marks

2024 NOV · Paper 01 · find part c in this mark scheme extract

10 (a)	$\left(3-9\left(\frac{4}{9}x+x^2\right)\right.$ or $\left.-9\left(-\frac{1}{3}+\frac{4}{9}x+x^2\right)\right)$ $3-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}\right]$ or $-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}-\frac{1}{3}\right]$ or $-1\left[9\left(x+\frac{2}{9}\right)^2-\frac{36}{81}-3\right]$ or $3+(1)\left[-9\left(x+\frac{2}{9}\right)^2+\frac{36}{81}\right]$ $\frac{31}{9}-9\left(x+\frac{2}{9}\right)^2$ $A=\frac{31}{9}$ $B=9$ $C=\frac{2}{9}$ oe	M1 A1 A1 A1 [4]
(b)	$"\frac{31}{9}"$	B1ft [1]
(c)	$\alpha+\beta=-\frac{4}{9}$ $\alpha\beta=-\frac{3}{9}$ oe $\frac{3\alpha}{\beta}+\frac{3\beta}{\alpha}\left(=\frac{3\left(\alpha^2+\beta^2\right)}{\alpha\beta}\right)=\frac{3\left((\alpha+\beta)^2-2\alpha\beta\right)}{\alpha\beta}$ $\frac{3\left[\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)\right]}{"-\frac{3}{9}"}$ $\left(=-\frac{70}{9}\right)$ $\left(\frac{3\alpha}{\beta}\times\frac{3\beta}{\alpha}=\right)9$ $x^2+"\frac{70}{9}"x+"9"\,(=0)$ $9x^2+70x+81=0$	B1 M1 A1ft B1 M1 A1cso [6]
(d)	$\left((x+y)^3=\right)x^3+3x^2y+3xy^2+y^3=x^3+y^3+3xy(x+y)$	B1*cso [1]
(e)	$(\alpha^2-\beta+\beta^2-\alpha=)$ $(\alpha+\beta)^2-2\alpha\beta-(\alpha+\beta)$ $\left((\alpha+\beta)^2=(\alpha+\beta)^2-2\alpha\beta\ =\frac{70}{81}\right)$ $\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)$ $\left(=\frac{106}{81}\right)$ **candidates may also transfer a value for $\alpha^2+\beta^2$ from working in part (c)** $\left[(\alpha^2-\beta)(\beta^2-\alpha)=\alpha^2\beta^2-\beta^3-\alpha^3+\alpha\beta\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha^3+\beta^3)$ or $(\alpha\beta)^2+\alpha\beta-\alpha^3-\beta^3$ $\left[(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ or $(\alpha\beta)^2+\alpha\beta-\left((\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right)$ $\left[=\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)\left(=\frac{226}{729}\right)\right]$ $"\frac{106}{81}"=-\frac{q}{3}$ or $"\frac{226}{729}"=\frac{r}{3}$ $q=-\frac{106}{27}$ and $r=\frac{226}{243}$	M1 A1ft M1 M1 dddM1 A1 [6]
		Total 18 marks

Type MS · Forming a new quadratic from the roots of another

Answer 16 of 22 · Question 5 part a · 2 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

5			
(a)	$x^2 - \frac{7}{2}x + 2 \ (=0)$	M1	
	$2x^2 - 7x + 4 = 0$	A1	(2)
(b)	$\frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$	B1	
	$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$	M1	
	$\frac{\frac{49}{4} - 4}{2}, = \frac{33}{8}$	dM1A1	
	$x^2 - \frac{33}{8}x + 1 \ (=0)$	M1	
	$8x^2 - 33x + 8 = 0$	A1	(6) **[8]**

(a)	
M1	Use $x^2 - (\text{sum of roots})x + \text{product of roots}$ (= 0 may be missing)
A1	Correct equation as shown or any **integer** multiple of this. Must have = 0 NB: A correct equation with no working scores 2
ALT	
M1	Eliminate α (or β) between the 2 equations and multiply through by α (or β)
A1	A correct quadratic equation with integer coefficients. Unknown can be α (or β)
NB;	isw any attempt to solve their equation.
(b)	
B1	Correct product of roots, seen explicitly or used.
M1	Attempt a single fraction for the sum of the roots with the numerator ready for substitution of known quantities. Denominator must be $\alpha\beta$.
dM1	Substitute numbers in their single fraction.
A1	Correct value for sum (as shown or equivalent fraction)
M1	Use $x^2 - (\text{sum of roots})x + \text{product of roots}$ (= 0 may be missing)
A1	Correct equation as shown or any **integer** multiple of this. Must have = 0

Type MS · Forming a new quadratic from the roots of another

Answer 16 of 22 · Question 5 part a · 2 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 17 of 22 · Question 5 part b · 6 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

5			
(a)	$x^2 - \frac{7}{2}x + 2 \ (=0)$	M1	
	$2x^2 - 7x + 4 = 0$	A1	(2)
(b)	$\frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$	B1	
	$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$	M1	
	$\frac{\frac{49}{4} - 4}{2}, = \frac{33}{8}$	dM1A1	
	$x^2 - \frac{33}{8}x + 1 \ (=0)$	M1	
	$8x^2 - 33x + 8 = 0$	A1	(6) **[8]**

(a)	
M1	Use $x^2 - (\text{sum of roots})x + \text{product of roots}$ (= 0 may be missing)
A1	Correct equation as shown or any **integer** multiple of this. Must have = 0 NB: A correct equation with no working scores 2
ALT	
M1	Eliminate α (or β) between the 2 equations and multiply through by α (or β)
A1	A correct quadratic equation with integer coefficients. Unknown can be α (or β)
NB;	isw any attempt to solve their equation.
(b)	
B1	Correct product of roots, seen explicitly or used.
M1	Attempt a single fraction for the sum of the roots with the numerator ready for substitution of known quantities. Denominator must be $\alpha\beta$.
dM1	Substitute numbers in their single fraction.
A1	Correct value for sum (as shown or equivalent fraction)
M1	Use $x^2 - (\text{sum of roots})x + \text{product of roots}$ (= 0 may be missing)
A1	Correct equation as shown or any **integer** multiple of this. Must have = 0

Type MS · Forming a new quadratic from the roots of another

Answer 17 of 22 · Question 5 part b · 6 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 18 of 22 · Question 8 part c · 8 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8 (a)	$\left(\frac{dy}{dx}=\right)4x-4$	M1
	Gradient of $l=8$	B1
	Gradient of normal $=-\frac{1}{8}$	B1
	$4x-4=-\frac{1}{8}\Rightarrow x=\frac{31}{32}$	M1A1 [5]
(b) (i)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)$	B1*cso [1]
(ii)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2=\left((\alpha+\beta)^2-2\alpha\beta\right)^2-2(\alpha\beta)^2$	B1*cso [1]
(c)	$\alpha+\beta=2\quad\alpha\beta=\frac{9}{2}$ oe	B1
	$\left(\alpha^3-\beta+\beta^3-\alpha=\alpha^3+\beta^3-(\alpha+\beta)\right)$	
	$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)-(\alpha+\beta)$	M1
	(sum of roots $=$) $2^3-3\left(\frac{9}{2}\right)(2)-2(=-21)$	A1
	$\left[\left(\alpha^3-\beta\right)\left(\beta^3-\alpha\right)=\right]\alpha^3\beta^3-\alpha^4-\beta^4+\alpha\beta$	M1
	$=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$	
	Now, $\left(\alpha^4+\beta^4\right)=\left((2)^2-2\times\frac{9}{2}\right)^2-2\times\left(\frac{9}{2}\right)^2=\left(-\frac{31}{2}\right)$	A1ft
	(product of roots $=$) $=\left(\frac{9}{2}\right)^3-\left(-\frac{31}{2}\right)+\frac{9}{2}\left(=\frac{889}{8}\right)$	A1
	$x^2-"-21"x+"\frac{889}{8}"$	M1
	$8x^2+168x+889=0$	A1 [8]
		Total 15 marks

Type MS · Forming a new quadratic from the roots of another

Answer 18 of 22 · Question 8 part c · 8 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
8 (a)	M1	For a correct differentiation.
	B1	For 8, identified as the gradient or used to calculate gradient of normal
	B1	For $-\frac{1}{8}$ stated as gradient of normal or used in subsequent work
	M1	For correctly solving their $ax+b=-\frac{1}{8}$
	A1	For $\frac{31}{32}$
(b) (i)	B1*cso	Complete and full algebra to show the given identity
(ii)	B1*cso	Complete and full algebra to show the given identity
(c)	B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
	M1	**Sum:** Using the result from part (b) and correctly manipulating to prepare for substitution of $\alpha+\beta$ **and** $\alpha\beta$
	A1	For -21
	M1	**Product:** For the correct algebraic expansion shown.
	A1ft	Correct substitution of their values into $\alpha^4+\beta^4$ within the product of roots
	A1	For $\frac{889}{8}$
	M1	Use $x^2-(\text{their sum of their roots})x+\text{their product of roots}$ "= 0" may be missing.
	A1	Correct equation oe with integer coefficients.

Type MS · Forming a new quadratic from the roots of another

Answer 18 of 22 · Question 8 part c · 8 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 19 of 22 · Question 6 part b · 5 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6(a)	$\alpha+\beta=\frac{3}{4}\quad \alpha\beta=-\frac{5}{4}$	B1
	$\frac{2\alpha}{\beta}+\frac{2\beta}{\alpha}=\frac{2\left(\alpha^2+\beta^2\right)}{\alpha\beta}=\frac{2\left(\left(\alpha+\beta\right)^2-2\alpha\beta\right)}{\alpha\beta}$	M1
	$=\frac{2\left(\frac{9}{16}+\frac{5}{2}\right)}{-\frac{5}{4}}=-\frac{49}{10}$	A1
	$\frac{2\alpha}{\beta}\times\frac{2\beta}{\alpha}=4$	B1
	$x^2+\frac{49}{10}x+4\ (=0)$	M1
	$10x^2+49x+40=0$	A1 (6)
(b)	$4\left(\alpha+\beta\right)=3=-\frac{p}{4}\quad p=-12$	M1A1
	$\left(3\alpha+\beta\right)\times\left(\alpha+3\beta\right)=3\left(\alpha^2+\beta^2\right)+10\alpha\beta$	M1
	$=3\left(\frac{9}{16}+\frac{5}{2}\right)-10\times\left(\frac{5}{4}\right)=-\frac{53}{16}$ oe	A1
	$\frac{q}{4}=-\frac{53}{16}\quad q=-\frac{53}{4}$ oe	A1 (5)
		[11]

(a)B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
M1	Attempt the sum of the roots of the new equation. Must reach a **correct** expression ready for substitution of values of $\alpha+\beta$ **and** $\alpha\beta$
A1	Correct value for the sum. Allow if $\alpha+\beta=-\frac{3}{4}$ has been used.
B1	Correct value for the product of roots of the new equation.
M1	Use $x^2-\left(\text{sum of roots}\right)x+\text{product of roots}$ "= 0" may be missing
A1	Correct final equation, as shown or an **integer** multiple of this. Must have = 0 now. Allow if $\alpha+\beta=-\frac{3}{4}$ has been used.
(b)	
M1	Use their value of $\alpha+\beta$ to obtain a value for the sum of the roots of $g\left(x\right)=0$ and equate to $\pm p/4$
A1	Obtain the correct value of p
M1	Attempt the product of the roots of $g\left(x\right)=0$ and obtain an expression ready for substitution of values. May use work from (a) for value of $\alpha^2+\beta^2$ so the expression shown is sufficient.
A1	Obtain the correct value for the product
A1	Correct value of q

Type MS · Forming a new quadratic from the roots of another

Answer 19 of 22 · Question 6 part b · 5 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$(x+1)\quad(4x-2)$	

Type MS · Forming a new quadratic from the roots of another

Answer 20 of 22 · Question 8 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Question number		
8 (a)	$\alpha+\beta=\frac{1}{3}\quad \alpha\beta=\frac{4}{3}$ $p=\alpha+\beta+\frac{\alpha+\beta}{\alpha\beta}$ $=\frac{1}{3}+\frac{\frac{1}{3}}{\frac{4}{3}}=\frac{7}{12}$ *	B1 M1 A1 cso* (3)
(b)	$q=\left(\alpha+\frac{1}{\alpha}\right)\left(\beta+\frac{1}{\beta}\right)=\alpha\beta+\frac{1}{\alpha\beta}+\frac{\beta^2+\alpha^2}{\alpha\beta}$ $=\alpha\beta+\frac{1}{\alpha\beta}+\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$ $=\frac{4}{3}+\frac{3}{4}+\frac{\frac{1}{9}-\frac{8}{3}}{\frac{4}{3}}=\frac{1}{6}$ oe	M1 A1 (M1 on ePen) dM1 A1 (4) **[7]**

Part	Mark	Additional Guidance
(a)	B1	$\alpha+\beta=\frac{1}{3},\ \alpha\beta=\frac{4}{3}$ can be explicit or implicitly used later
	M1	For an initial expression correctly adding the roots of g(x) and an attempt to simplify. **Minimum attempt must involve terms $\alpha+\beta$ and an attempt to simplify, bringing the fraction part to a common denominator $\alpha\beta$ written as a single fraction.**
	A1*	cso*
(b)	M1	For multiplying the roots of g(x). **Minimum attempt must involve a correct multiplication of the brackets with terms $\alpha\beta+\frac{1}{\alpha\beta}$ and an attempt to bring the fraction part to a common denominator $\alpha\beta$**
	A1 (M1 ePen)	Correct expression as shown.
	dM1	Substitutes their values for $\alpha+\beta$ and $\alpha\beta$ **correctly** into their expression (this expression must be ready for substitution of **both** $\alpha+\beta$ **and** $\alpha\beta$)
	A1	$\frac{1}{6}$ oe

Type MS · Forming a new quadratic from the roots of another

Answer 20 of 22 · Question 8 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 21 of 22 · Question 8 part b · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

8 (a)	$\alpha+\beta=\frac{1}{3}\quad \alpha\beta=\frac{4}{3}$	B1
	$p=\alpha+\beta+\frac{\alpha+\beta}{\alpha\beta}$	M1
	$=\frac{1}{3}+\frac{\frac{1}{3}}{\frac{4}{3}}=\frac{7}{12}$ *	A1 cso* (3)
(b)	$q=\left(\alpha+\frac{1}{\alpha}\right)\left(\beta+\frac{1}{\beta}\right)=\alpha\beta+\frac{1}{\alpha\beta}+\frac{\beta^2+\alpha^2}{\alpha\beta}$	M1
	$=\alpha\beta+\frac{1}{\alpha\beta}+\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$	A1 (M1 on ePen)
	$=\frac{4}{3}+\frac{3}{4}+\frac{\frac{1}{9}-\frac{8}{3}}{\frac{4}{3}}=\frac{1}{6}$ oe	dM1 A1 (4) **[7]**

Part	Mark	Additional Guidance
(a)	B1	$\alpha+\beta=\frac{1}{3},\ \alpha\beta=\frac{4}{3}$ can be explicit or implicitly used later
	M1	For an initial expression correctly adding the roots of g(x) and an attempt to simplify. **Minimum attempt must involve terms $\alpha+\beta$ and an attempt to simplify, bringing the fraction part to a common denominator $\alpha\beta$ written as a single fraction.**
	A1*	cso*
(b)	M1	For multiplying the roots of g(x). **Minimum attempt must involve a correct multiplication of the brackets with terms $\alpha\beta+\frac{1}{\alpha\beta}$ and an attempt to bring the fraction part to a common denominator $\alpha\beta$**
	A1 (M1 ePen)	Correct expression as shown.
	dM1	Substitutes their values for $\alpha+\beta$ and $\alpha\beta$ **correctly** into their expression (this expression must be ready for substitution of **both** $\alpha+\beta$ **and** $\alpha\beta$)
	A1	$\frac{1}{6}$ oe

Type MS · Forming a new quadratic from the roots of another

Answer 21 of 22 · Question 8 part b · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Forming a new quadratic from the roots of another

Answer 22 of 22 · Question 10 part e · 6 marks

2024 NOV · Paper 01 · find part e in this mark scheme extract

10 (a)	$\left(3-9\left(\frac{4}{9}x+x^2\right)\right.$ or $\left.-9\left(-\frac{1}{3}+\frac{4}{9}x+x^2\right)\right)$ $3-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}\right]$ or $-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}-\frac{1}{3}\right]$ or $-1\left[9\left(x+\frac{2}{9}\right)^2-\frac{36}{81}-3\right]$ or $3+(1)\left[-9\left(x+\frac{2}{9}\right)^2+\frac{36}{81}\right]$ $\frac{31}{9}-9\left(x+\frac{2}{9}\right)^2$ $A=\frac{31}{9}$ $B=9$ $C=\frac{2}{9}$ oe	M1 A1 A1 A1 [4]
(b)	$"\frac{31}{9}"$	B1ft [1]
(c)	$\alpha+\beta=-\frac{4}{9}$ $\alpha\beta=-\frac{3}{9}$ oe $\frac{3\alpha}{\beta}+\frac{3\beta}{\alpha}\left(=\frac{3\left(\alpha^2+\beta^2\right)}{\alpha\beta}\right)=\frac{3\left((\alpha+\beta)^2-2\alpha\beta\right)}{\alpha\beta}$ $\frac{3\left[\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)\right]}{"-\frac{3}{9}"}$ $\left(=-\frac{70}{9}\right)$ $\left(\frac{3\alpha}{\beta}\times\frac{3\beta}{\alpha}=\right)9$ $x^2+"\frac{70}{9}"x+"9"\,(=0)$ $9x^2+70x+81=0$	B1 M1 A1ft B1 M1 A1cso [6]
(d)	$\left((x+y)^3=\right)x^3+3x^2y+3xy^2+y^3=x^3+y^3+3xy(x+y)$	B1*cso [1]
(e)	$(\alpha^2-\beta+\beta^2-\alpha=)$ $(\alpha+\beta)^2-2\alpha\beta-(\alpha+\beta)$ $\left((\alpha+\beta)^2=(\alpha+\beta)^2-2\alpha\beta=\frac{70}{81}\right)$ $\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)$ $\left(=\frac{106}{81}\right)$ **candidates may also transfer a value for $\alpha^2+\beta^2$ from working in part (c)** $\left[(\alpha^2-\beta)(\beta^2-\alpha)=\alpha^2\beta^2-\beta^3-\alpha^3+\alpha\beta\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha^3+\beta^3)$ or $(\alpha\beta)^2+\alpha\beta-\alpha^3-\beta^3$ $\left[(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ or $(\alpha\beta)^2+\alpha\beta-\left((\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right)$ $\left[=\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)\left(=\frac{226}{729}\right)\right]$ $"\frac{106}{81}"=-\frac{q}{3}$ or $"\frac{226}{729}"=\frac{r}{3}$ $q=-\frac{106}{27}$ and $r=\frac{226}{243}$	M1 A1ft M1 M1 dddM1 A1 [6]
		Total 18 marks